THE STRUCTURE SHEAF OF THE LISSE-ETALE AND FLAT-FPPF SITES PUSHES FORWARD TO THE STRUCTURE SHEAF

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- The g_! functor. Modules on Sites, Definition sites-modules-definition-g-shriek. Colimit presheaf formula for g_!O'.
- Etale dominates smooth. Topologies on Spaces, Lemma spaces-topologies-lemma-etale-dominates-smooth. Lets V be replaced so that V -> U exists.

Source reference. Original lemma 0788: sources/stacks-chow/tags.txt, source lines 9228–9228. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0789: sources/stacks-chow/tags.txt, source lines 9229–9229. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04BF: sources/stacks-chow/tags.txt, source lines 5650–5650. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0CFV: sources/stacks-chow/tags.txt, source lines 15625–15625. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. Notation as in Lemmas 0788 and 0789.*

(1) *We have $g_!\mathcal{O}_{\mathcal{X}_{lisse,étale}} = \mathcal{O}_{\mathcal{X}}$.*
(2) *We have $g_!\mathcal{O}_{\mathcal{X}_{flat,fppf}} = \mathcal{O}_{\mathcal{X}}$.*

Proof. In this proof we write $\mathcal{C} = \mathcal{X}_{étale}$ (resp. $\mathcal{C} = \mathcal{X}_{fppf}$) and we denote $\mathcal{C}' = \mathcal{X}_{lisse,étale}$ (resp. $\mathcal{C}' = \mathcal{X}_{flat,fppf}$). Then $\mathcal{C}'$ is a full subcategory of $\mathcal{C}$. In this proof we will think of objects V of $\mathcal{C}$ as schemes over $\mathcal{X}$ and objects U of $\mathcal{C}'$ as schemes smooth (resp. flat) over $\mathcal{X}$. Finally, we write $\mathcal{O} = \mathcal{O}_{\mathcal{X}}$ and $\mathcal{O}' = \mathcal{O}_{\mathcal{X}_{lisse,étale}}$ (resp. $\mathcal{O}' = \mathcal{O}_{\mathcal{X}_{flat,fppf}}$). In the notation above we have $\mathcal{O}(V) = \Gamma(V, \mathcal{O}_V)$ and $\mathcal{O}'(U) = \Gamma(U, \mathcal{O}_U)$. Consider the $\mathcal{O}$-module homomorphism $g_!\mathcal{O}' \to \mathcal{O}$ adjoint to the identification $\mathcal{O}' = g^{-1}\mathcal{O}$.

Recall that $g_!\mathcal{O}'$ is the sheaf associated to the presheaf $g_{p!}\mathcal{O}'$ given by the rule

$$V \longmapsto \operatorname{colim}_{V \to U} \mathcal{O}'(U)$$

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where the colimit is taken in the category of abelian groups (Modules on Sites, Definition 04BF). Below we will use frequently that if

$$V \to U \to U'$$

are morphisms and if $f' \in \mathcal{O}'(U')$ restricts to $f \in \mathcal{O}'(U)$, then $(V \to U, f)$ and $(V \to U', f')$ define the same element of the colimit. Also, $g_!\mathcal{O}' \to \mathcal{O}$ maps the element $(V \to U, f)$ simply to the pullback of f to V.

Let us prove that $g_!\mathcal{O}' \to \mathcal{O}$ is surjective. Let $h \in \mathcal{O}(V)$ for some object V of $\mathcal{C}$. It suffices to show that h is locally in the image. Choose an object U of $\mathcal{C}'$ corresponding to a surjective smooth morphism $U \to \mathcal{X}$. Since $U \times_{\mathcal{X}} V \to V$ is surjective smooth, after replacing V by the members of an étale covering of V we may assume there exists a morphism $V \to U$, see Topologies on Spaces, Lemma 0CFV. Using h we obtain a morphism $V \to U \times \mathbf{A}^1$ such that writing $\mathbf{A}^1 = \mathrm{Spec}(\mathbf{Z}[t])$ the element $t \in \mathcal{O}(U \times \mathbf{A}^1)$ pulls back to h. Since $U \times \mathbf{A}^1$ is an object of $\mathcal{C}'$ we see that $(V \to U \times \mathbf{A}^1, t)$ is an element of the colimit above which maps to $h \in \mathcal{O}(V)$ as desired.

Suppose that $s \in g_!\mathcal{O}'(V)$ is a section mapping to zero in $\mathcal{O}(V)$. To finish the proof we have to show that s is zero. After replacing V by the members of a covering we may assume s is an element of the colimit

$$\mathrm{colim}_{V \to U} \mathcal{O}'(U)$$

Say $s = \sum(\varphi_i, s_i)$ is a finite sum with $\varphi_i : V \to U_i$, U_i smooth (resp. flat) over $\mathcal{X}$, and $s_i \in \Gamma(U_i, \mathcal{O}_{U_i})$. Choose a scheme W surjective étale over the algebraic space $U = U_1 \times_{\mathcal{X}} \ldots \times_{\mathcal{X}} U_n$. Note that W is still smooth (resp. flat) over $\mathcal{X}$, i.e., defines an object of $\mathcal{C}'$. The fibre product

$$V' = V \times_{(\varphi_1, \ldots, \varphi_n), U} W$$

is surjective étale over V, hence it suffices to show that s maps to zero in $g_!\mathcal{O}'(V')$. Note that the restriction $\sum(\varphi_i, s_i)|_{V'}$ corresponds to the sum of the pullbacks of the functions s_i to W. In other words, we have reduced to the case of (φ, s) where $\varphi : V \to U$ is a morphism with U in $\mathcal{C}'$ and $s \in \mathcal{O}'(U)$ restricts to zero in $\mathcal{O}(V)$. By the commutative diagram

$$\begin{array}{ccc} V & \xrightarrow{} & U \times \mathbf{A}^1 \\ & \searrow^{\varphi} & \uparrow{\scriptstyle (\mathrm{id},0)} \\ & & U \end{array} \qquad (\varphi,0)$$

we see that $((\varphi, 0) : V \to U \times \mathbf{A}^1, \mathrm{pr}_2^* x)$ represents zero in the colimit above. Hence we may replace U by $U \times \mathbf{A}^1$, φ by $(\varphi, 0)$ and s by $\mathrm{pr}_1^* s + \mathrm{pr}_2^* x$. Thus we may assume that the vanishing locus $Z : s = 0$ in U of s is smooth (resp. flat) over $\mathcal{X}$. Then we see that $(V \to Z, 0)$ and (φ, s) have the same value in the colimit, i.e., we see that the element s is zero as desired. □